

2026S1PH1012: Physics A
Vectors and Circular Motion

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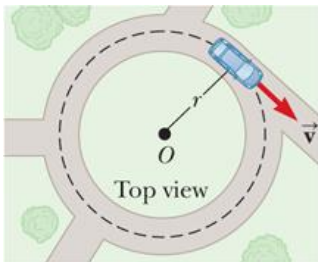
Week 5
Serway Chap 4.4 – 4.5

"We are what we repeatedly do; excellence, then, is not an act but a habit."

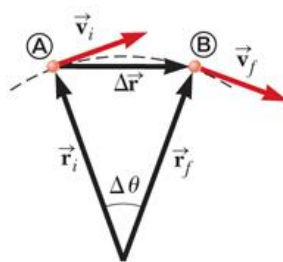
- Aristotle (384 BC – 322 BC)

Self-check – after completing the lessons (lecture and tutorial) for this topic, are you able to

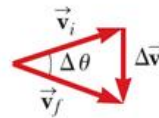
1. recognize circumstances where vector subtraction is needed to solve problems and apply it
2. compute instantaneous change in velocity of objects moving in 2D or 3D
3. solve problems related to objects moving in regular circular motion
4. determine the acceleration for objects moving in a curve
5. determine forces and centripetal objects moving in horizontal, vertical and conical circular motion
6. solve problems related to objects moving in circular motion with constant angular acceleration, including determining tangential acceleration and centripetal acceleration



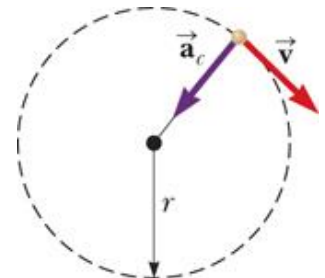
a



b



c



Serway Fig 4.16

A car moving at constant speed in a uniform circle undergoes velocity change

Serway: Centripetal Acceleration

Kinematics in 2D – Vector Subtraction to compute Vector Change

In two or three dimensions, the displacement is a vector given by:

$$\Delta \vec{r} = \vec{r}_2 - \vec{r}_1$$

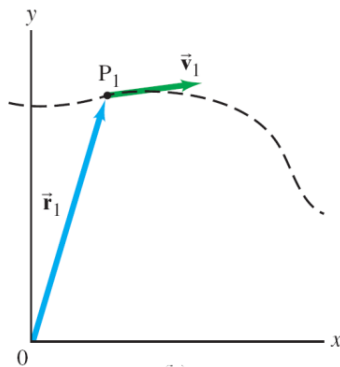
To compute average velocity, we compute

$$\vec{v}_{avg} = \frac{\Delta \vec{r}}{\Delta t}.$$

$$Avg\ Vel = \frac{\text{change in position (or displacement)}}{\text{time}}$$

As $\Delta \vec{r}$ and Δt becomes smaller and smaller, in the limit where $\Delta t \rightarrow 0$ the average velocity becomes the instantaneous velocity:

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$



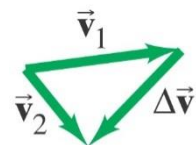
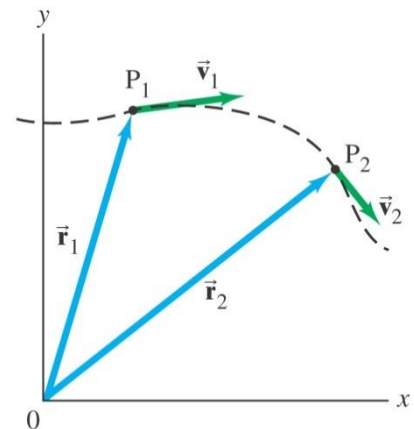
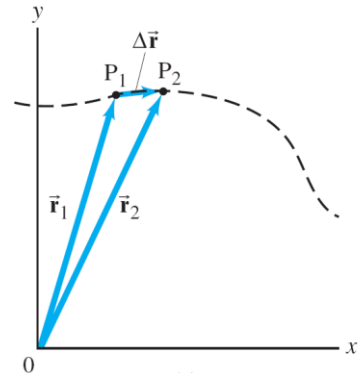
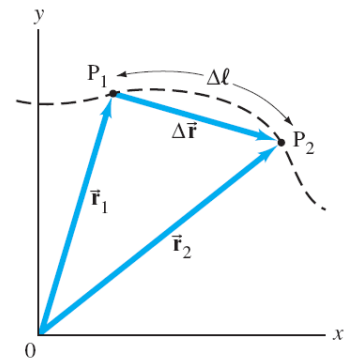
Similarly, average acceleration and instantaneous acceleration can be computed. To compute average acceleration,

$$\vec{a}_{avg} = \frac{\Delta \vec{v}}{\Delta t}.$$

$$Avg\ Acc = \frac{\text{change in velocity}}{\text{time}}$$

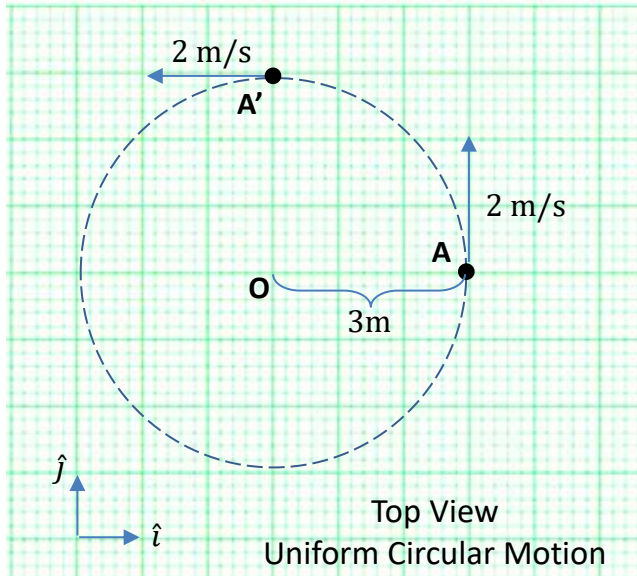
As $\Delta \vec{v}$ and Δt becomes smaller and smaller, in the limit where $\Delta t \rightarrow 0$ the average acceleration becomes the instantaneous acceleration:

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$



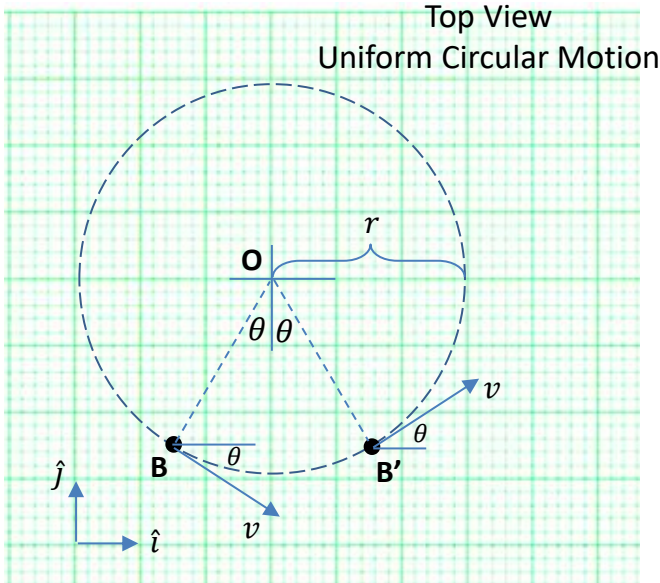
Velocity change for object moving in a circle

Take O as the center of the circle and also the origin of the x-y coordinate system. A particle is moving at a **constant speed** of 2 m/s round a circle of radius 3m.



- Write down the position of the particle when it is at point A and A'.
 - What is the displacement of the particle when it moves from A to A'?
 - What is the angular displacement (in degrees and radians) of the particle when it moves from A to A'?
 - How much time does the particle take to get from A to A'?
-
- What is the angular velocity of the particle when it moves from A to A'?
 - What is the instantaneous velocity of the particle at A and A'?
 - What is the change in velocity of the particle when it moves from A to A'?
 - What is the average acceleration of particle when it moves from A to A'?
 - Is the speed of the particle constant?
 - Is the velocity of the particle constant?
 - Is there a horizontal force acting on the particle when it moves in a circle?

Instantaneous velocity change and acceleration for object moving in a circle



Take O as the center of the circle and also the origin of the x-y coordinate system.

- How much time does the particle take to get from B to B'?
- What is the instantaneous velocity of the particle at B and B'?

- What is the average acceleration of the particle when it moves from B to B'?

Average acceleration,

$$\bar{a} = \frac{v_f - v_i}{t_f - t_i} = \frac{\Delta v}{\Delta t}$$

Instantaneous acceleration,

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$

θ (in rad)	0.1	0.01	0.001
$\frac{\sin \theta}{\theta}$			

- Taking the limit where $\Delta t \rightarrow 0$ and hence $\theta \rightarrow 0$, determine the instantaneous acceleration of the particle.

Circular Motion

Circular Measures

One **radian** is defined as the angle subtended by an arc whose length is equal to the radius R .

The arc length l in the figure is given by $l = R\theta$ where θ is in radians (rad).

The circumference of a circle C is given by $C = 2\pi R$. A complete circle is 360° , hence

$$360^\circ \equiv 2\pi \text{ rad}$$

$$1 \text{ rad} \equiv \frac{360^\circ}{2\pi} \approx 57.3^\circ$$

Angular Displacement / Velocity

For translational motion, we have linear displacement. For rotational motion, we consider angular displacement. In the figure, the wheel rotates from initial position θ_1 to final position θ_2 , thus the angular displacement is given by

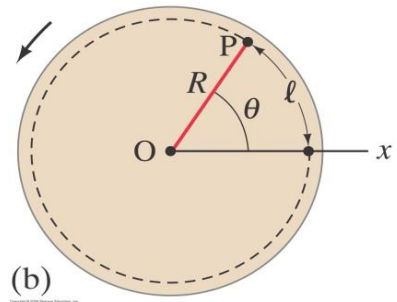
$$\Delta\theta = \theta_2 - \theta_1.$$

Similar to average linear velocity, average angular velocity is defined by

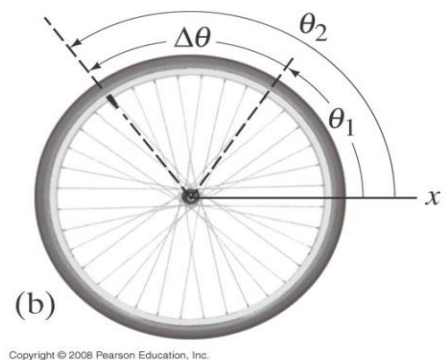
$$\bar{\omega} = \frac{\Delta\theta}{\Delta t}.$$

The instantaneous angular velocity is defined by

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t}$$

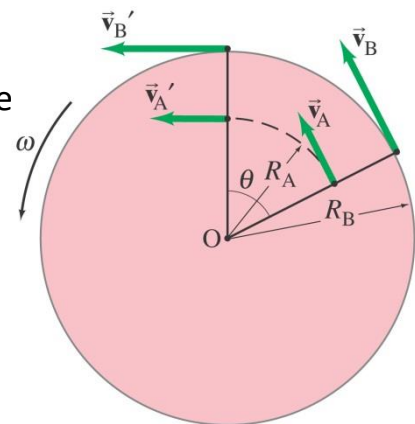


Giancoli Fig 10.2: Arc length



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Giancoli Fig 10.4: Angular displacement



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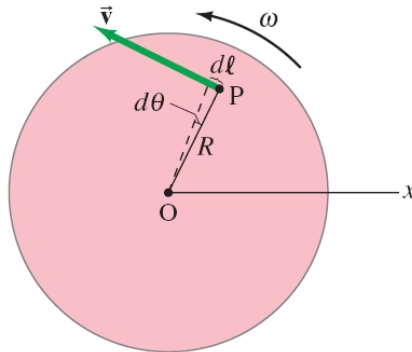
Giancoli Fig 10.6: Tangential velocity

Circular Motion (Giancoli Chap 10.1)

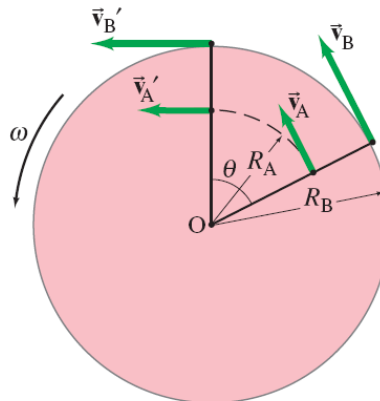
Every point on a rotating body has an angular velocity ω and a linear velocity v .

They are related:

$$v = R\omega$$



Objects farther from the axis of rotation will move faster.



Recall that earlier that we learnt the definition of acceleration in a 1-D context

Average acceleration,

$$\bar{a} = \frac{v_f - v_i}{t_f - t_i} = \frac{\Delta v}{\Delta t}$$

Instantaneous acceleration,

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$

Now, we generalize to 2-D and higher D by adding $\vec{}$ and see how the meaning changes

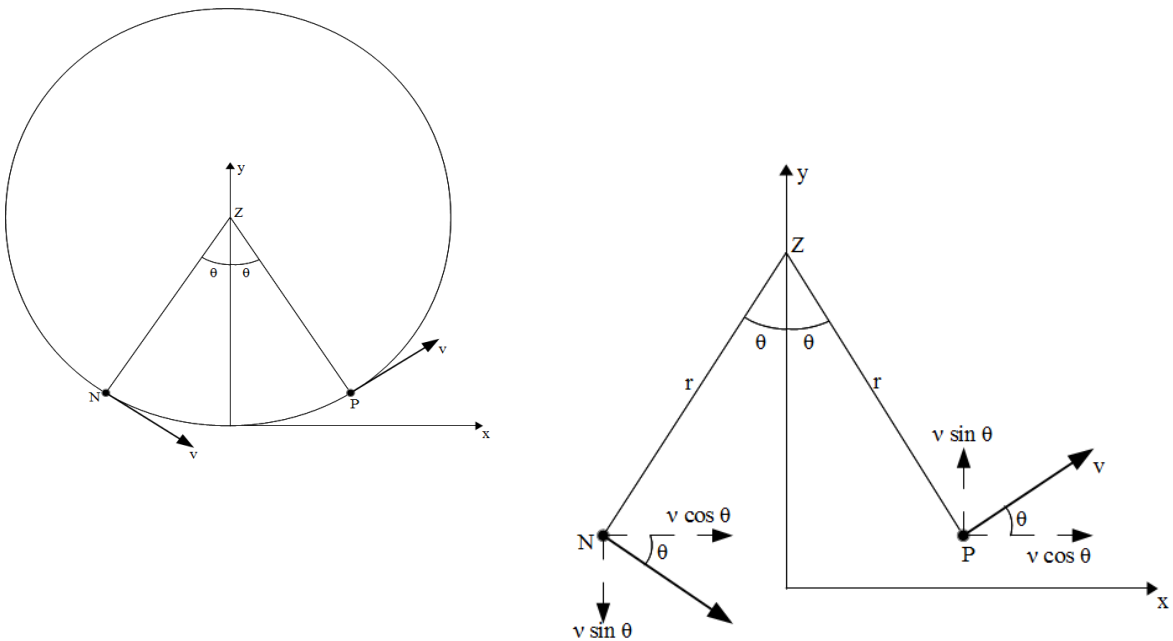
Average acceleration,

$$\bar{\vec{a}} = \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i} = \frac{\Delta \vec{v}}{\Delta t}$$

Instantaneous acceleration,

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

Instantaneous Acceleration in Uniform Circular Motion



When $\theta \approx 0$, points N and P are almost at the same place



The velocities before and after are almost parallel. The change in velocity

$$\Delta \vec{v} = \vec{v}_f - \vec{v}_i$$

is almost perpendicular to both $\vec{v}_f$ and $\vec{v}_i$.

In the limit, $\Delta t \rightarrow 0$, $\theta \rightarrow 0$ and $\Delta \vec{v} \perp \vec{v}$ with $\Delta \vec{v} = 2v \sin \theta$

Thus, the instantaneous acceleration

$$\begin{aligned} a &= \lim_{\theta \rightarrow 0} \frac{\Delta v}{\Delta t} = \lim_{\theta \rightarrow 0} \frac{2v \sin \theta}{\frac{2r\theta}{v}} \\ &= \lim_{\theta \rightarrow 0} \frac{v^2 \sin \theta}{r \theta} = \frac{v^2}{r}. \end{aligned}$$

This acceleration that is always perpendicular to the instantaneous velocity and pointing towards the center of the circle is known as the **centripetal acceleration**.

Since $v = r\omega$, we can also write $a_c = \frac{v^2}{r} = \frac{(r\omega)^2}{r} = r\omega^2$.

Uniform Circular Motion

Giancoli pg 276 Problem 10.8

A rotating merry-go-round makes one complete revolution in 4.0 s. (a) What is the angular speed of a child seated 1.2 m from the center? (b) What is the linear speed of a child seated 1.2 m from the center? Determine the centripetal acceleration experienced by the child.

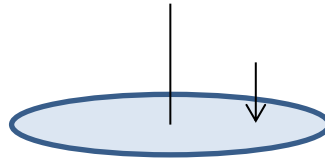


<https://www.youtube.com/watch?v=70NH2F7RBjs>

EXERCISE Giancoli Prob 10.5 (a) A grinding wheel 0.35 m in diameter rotates at 2500 rpm. Calculate its angular velocity in rad/s (b) What are the linear speed of a point on the edge of the grinding wheel?

Giancoli pg 253 Example 10-4: Hard drive

The platter of the hard drive of a computer rotates at 7200 rpm (rpm = revolutions per minute = rev/min). (a) What is the angular velocity (rad/s) of the platter? (b) If the reading head of the drive is located 3.00 cm from the rotation axis, what is the linear speed of the point on the platter just below it? (c) If a single bit requires $0.50 \mu\text{m}$ of length along the direction of motion, how many bits per second can the writing head write when it is 3.00 cm from the axis?



a) $\omega =$

b) $v = R\omega = (3.00 \times 10^{-2}\text{m}) \left(754 \frac{\text{rad}}{\text{s}} \right) = 22.6 \frac{\text{m}}{\text{s}}$

c) Each bit requires $0.50 \times 10^{-6}\text{m}$, so that's 22.6 m/s .

$$\text{Bits passing per head} = \frac{22.6 \frac{\text{m}}{\text{s}}}{0.50 \times 10^{-6} \text{m/bit}} = 45 \times 10^6 \text{ bits per s}$$

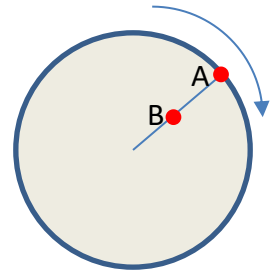


<https://www.youtube.com/watch?v=3owqvmMf6No>

Uniform Circular Motion.

Question: Two points are marked on a circular disc. The disc is moving with uniform speed and the diagram on the right shows the top view. Point A is further away from the center of the disc as compared to point B.

- a) Which point has a larger angular velocity?
- b) Which point has a larger linear velocity?
- c) Which point has a larger centripetal acceleration?



Top view

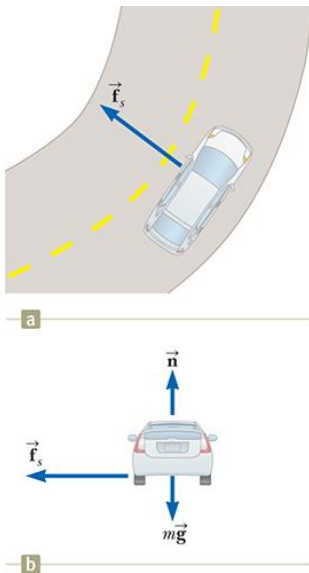
EXERCISE Giancoli Prob 10.10 Calculate the angular velocity of the Earth about its axis. Given that the radius of the Earth at the equator is about 6380 km, calculate the linear velocity of someone at the equator. Calculate also the centripetal acceleration. How does it compare to someone in London?



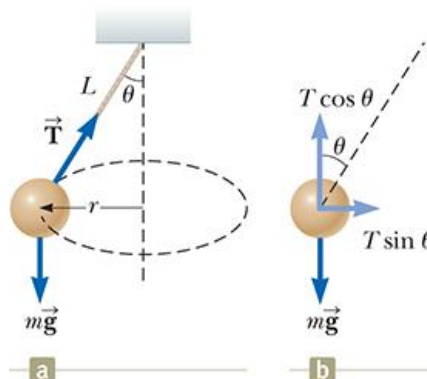
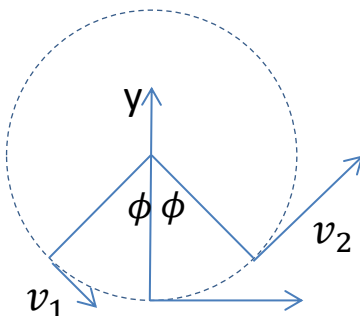
"In theory, there is no difference between theory and practice; In practice, there is."

- Chuck Reid

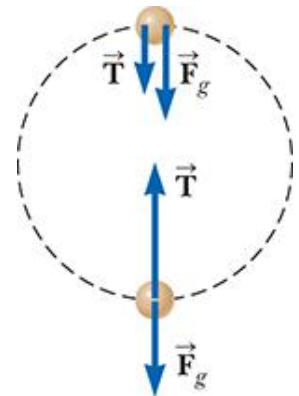
Uniform and Non-uniform Circular Motion



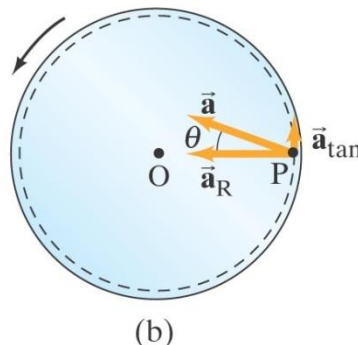
Serway Fig 6.4: Static friction providing centripetal force for car moving in circular arc



Serway Fig 6.3: Conical Pendulum



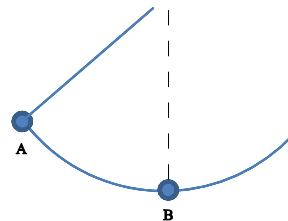
Serway Fig 6.31: Forces acting on an object moving in a vertical circle



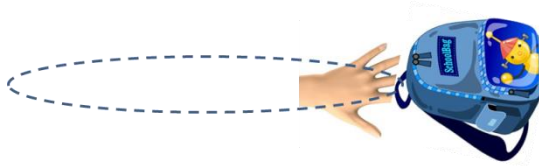
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Giancoli pg 252 Fig 10.8b

Object speeding up round in a circle



Swinging School Bag



A kid is swinging his school bag round in a horizontal circle at constant speed.

- If he wants to swing the bag even faster at the same radius, does he have to exert a larger, smaller or the same force F_T as compared to before?
- If he wants to swing the bag in a smaller circle but with the same number of rounds per second, does he have to exert a larger, smaller or the same force F_T as compared to before?

Centripetal Force

For forces acting on an object of mass m along a straight line, we can apply Newton's second law and write

$$\sum \vec{F} = m\vec{a}$$

For motion in a circle, the net radial forces (along the radius), that keeps the object moving a circular arc (deviating from a straight line) is known as the centripetal force

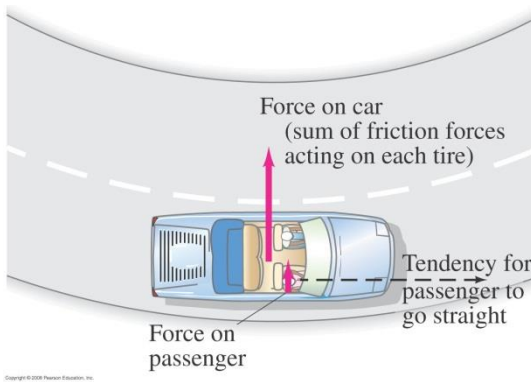
$$\text{Along radial direction, } F_c = \sum F = \frac{mv^2}{r} = mr\omega^2$$

1. Basic concepts in circular motion (Dynamics)

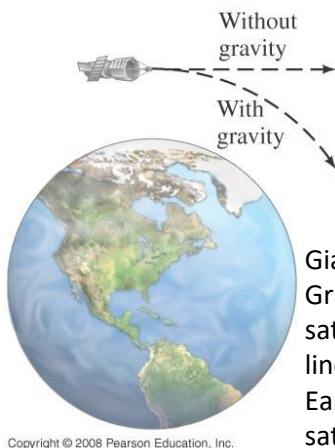
Newton's First Law states that

An object at rest remains at rest and an object in motion will continue in motion with a constant velocity (i.e., constant speed in a straight line) unless it experiences a non-zero net external force.

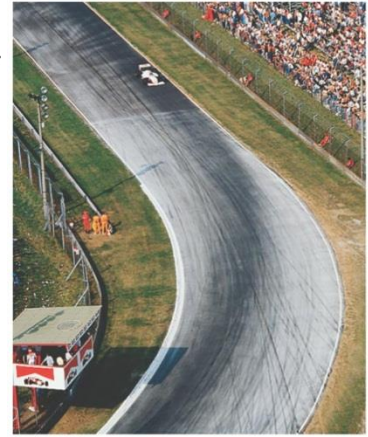
Therefore, when an object moves in a circle or part of a circular arc, there is a non-zero net external force acting on the object. The force that causes the object to deviate from a straight linear path to move in a circular arc is known as the centripetal force. In uniform circular motion, we can derive the centripetal acceleration a_c by computing the rate of change of velocity.



Giancoli pg 126 Fig 5.21:
Forces acting on a car and its passengers to keep them going round a circular arc



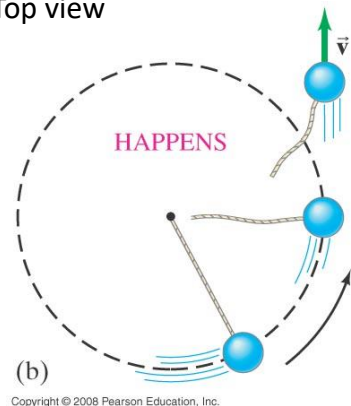
Giancoli pg 146 Fig 6.12:
Gravitational force causes the satellite to deviate from a straight line. At the right distance from Earth's centre and velocity, the satellite moves in a circular path.



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Giancoli pg 126 Fig 5.22:
Note skid tracks left by racing cars that failed to go round the bend.

Top view



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Giancoli pg 123 Fig 5.16 (b):
Tension in a string keeps the ball moving in a circle, when it breaks, the ball flies off tangentially.

Giancoli pg 126-27 Example 5-14: Skidding on a curve

A 1000-kg car rounds a curve on a flat road of radius 50 m at a speed of 15 m/s (54 km/h). Will the car follow the curve, or will it skid? Assume: (a) the pavement is dry and the coefficient of static friction is $\mu_s = 0.60$;

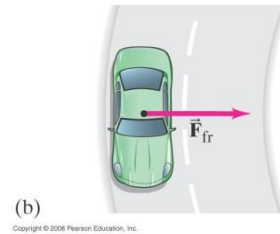
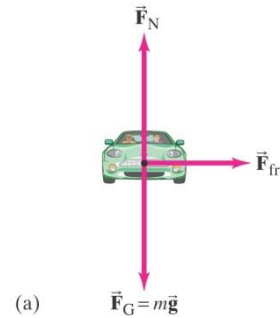
(b) the pavement is icy and $\mu_s = 0.25$.

$$F_N - mg = 0$$

$$\Rightarrow F_N = 1000 \times 9.8 = 9800 \text{ N}$$

a) Dry: $(F_s)_{\max} = \mu_s F_N =$

b) Icy: $(F_s)_{\max} = \mu_s F_N =$

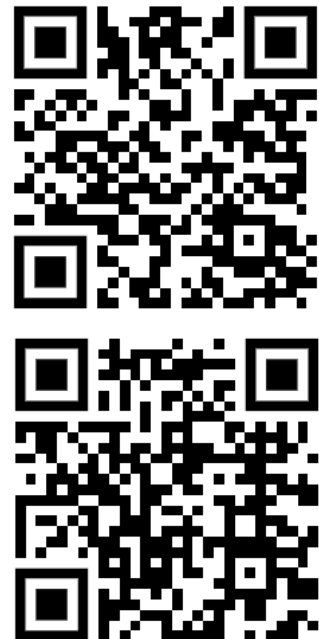


Giancoli pg 126 Fig 5.23

<https://www.youtube.com/watch?v=eGZWVwcaq0U>

Indoor Cycling:

https://www.youtube.com/watch?v=wgHnEXI_zug



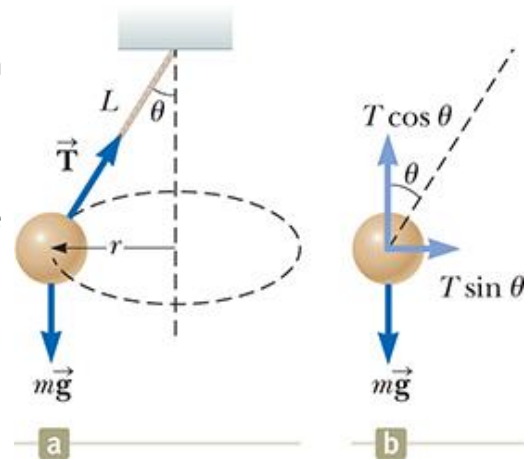
Serway Example 6-1: Conical Pendulum (uniform Circular Motion)

A small ball of mass m is suspended from a string of length L . The ball revolves with constant speed v in a horizontal circle of radius r as shown in the figure. (Because the string sweeps out the surface of a cone, the system is known as a *conical pendulum*.) Find an expression for v in terms of the length of the string and the angle it makes with the vertical in the figure.

Analysis of forces:

$$\text{vertical: } T \cos \theta - mg = ma = 0 \quad (1)$$

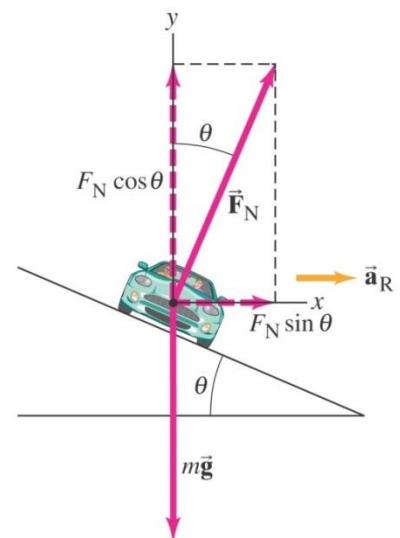
$$\text{horizontal: } T \sin \theta = \frac{mv^2}{r} \quad (2)$$



Serway Fig 6.3: Conical Pendulum

Giancoli pg 127 Example 5-15: Banking angle

(a) For a car traveling with speed v around a curve of radius r , determine a formula for the angle at which a road should be banked so that no friction is required. (b) What is this angle for an expressway off-ramp curve of radius 50 m at a design speed of 50 km/h (13.9 m/s)?



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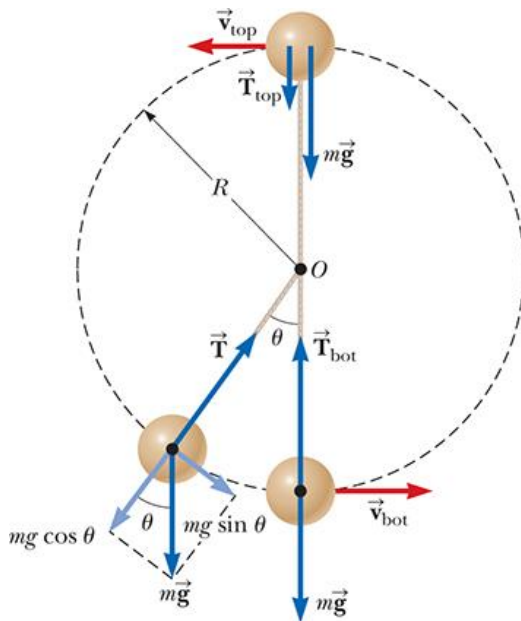
Giancoli pg 127 Fig 5.24

Note: Banking angle is independent of mass of vehicle

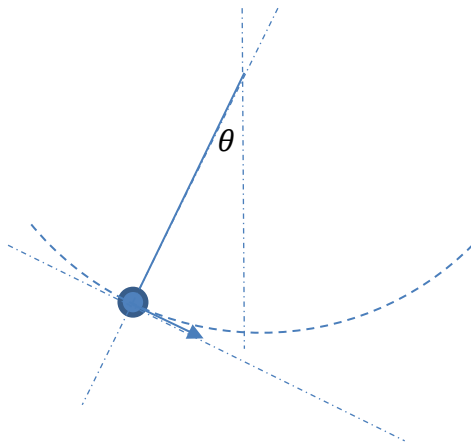
Non-Uniform Circular Motion

Complete Vertical Circular Motion

An analysis of forces



Serway Fig 6.9: Forces acting on a ball connected to a cord going in a vertical circle



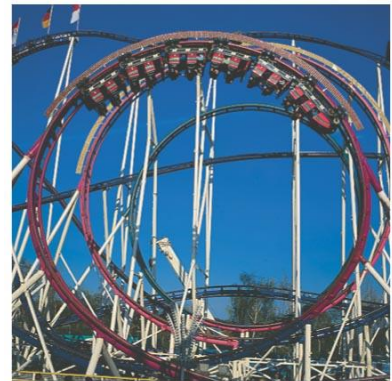
EXERCISE

Example [G5.55] Tarzan plans to cross a gorge by swinging in an arc from a hanging vine as shown. If his arms are capable of exerting a force of 1350 N on the rope, what is the maximum speed he can tolerate at the lowest point of his swing? His mass is 78 kg and the vine is 5.2 m long.



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Example [G5.40] At what minimum speed must a roller coaster be traveling when upside down at the top of a circle so that the passengers do not fall out? Assume a radius of curvature of 7.6 m.



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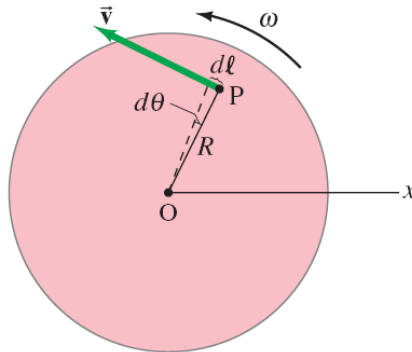
<https://www.youtube.com/watch?v=Chlf6NsfJ1I>

General Circular Motion (Giancoli Chap 10.1) with constant tangential acceleration (magnitude)

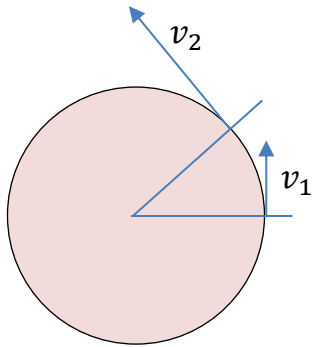
Every point on a rotating body has an angular velocity ω and a linear velocity v .

They are related:

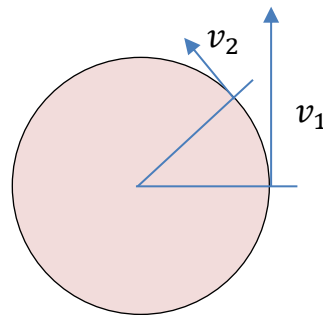
$$v = R\omega$$



In general, the angular velocity does not have to be constant.



Increasing ω and v .



Decreasing ω and v .

If the angular velocity of a rotating object changes, it has a tangential acceleration:

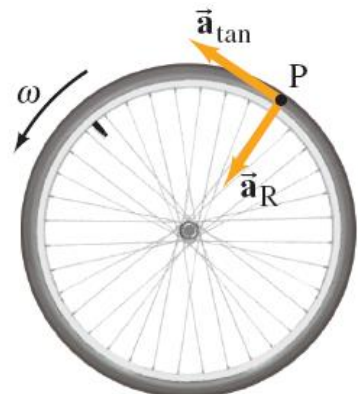
$$a_{tan} = \frac{dv}{dt} = R \frac{d\omega}{dt} = R\alpha$$

Even if the angular velocity is constant, each point on the object has a centripetal acceleration (to discuss later):

$$a_R = \frac{v^2}{R} = \frac{(R\omega)^2}{R} = R\omega^2$$

The total linear acceleration at a point at any instant is the vector

$$\vec{a} = \vec{a}_{tan} + \vec{a}_R$$



Giancoli pg 276, Problem 10.4

The blades of a blender rotate at a rate of 6500 rpm (rev per min). When the motor is turned off during the operation, the blades slowed down to rest in 4.0 s. What is the angular acceleration as the blades slow down?

Angular vs Linear displacement / velocity / acceleration

Angular	Linear
θ	$s = R\theta$
ω	$v_{tan} = R\omega$
α	$a_{tan} = R\alpha$
$\omega_f = \omega_i + \alpha t$	$v_f = v_i + at$
	$s = v_i t + \frac{1}{2}at^2$
	$v_f^2 = v_i^2 + 2as$
	$s = \frac{1}{2}(v_f + v_i)t$

Radial (Centripetal) Acceleration

$$a_r = \frac{v^2}{r} = r\omega^2$$

Giancoli pg 252, Example 10-3

A carousel is initially at rest. At $t = 0$ it is given a constant angular acceleration $\alpha = 0.060 \text{ rad/s}^2$, which increases its angular velocity for 8.0 s . At $t = 8.0 \text{ s}$, determine the magnitude of the following quantities: (a) the angular velocity of the carousel; (b) the linear velocity of a child located 2.5 m from the center; (c) the tangential (linear) acceleration of that child; (d) the centripetal acceleration of the child; and (e) the total linear acceleration of the child.

$$\begin{aligned} \text{a) } \omega_2 &= \omega_1 + \alpha \Delta t \\ &= 0 + \left(0.060 \frac{\text{rad}}{\text{s}^2}\right)(8.0 \text{ s}) = 0.48 \frac{\text{rad}}{\text{s}}. \end{aligned}$$

$$\text{b) } v = R\omega = (2.5 \text{ m})\left(0.49 \frac{\text{rad}}{\text{s}}\right) = 1.2 \frac{\text{m}}{\text{s}}$$

$$\text{c) } a_{\text{tan}} = R\alpha = (2.5 \text{ m})\left(0.060 \frac{\text{rad}}{\text{s}^2}\right) = 0.15 \frac{\text{m}}{\text{s}^2}$$

$$\text{d) } a_R = \frac{v^2}{R} = \frac{\left(1.2 \frac{\text{m}}{\text{s}}\right)^2}{(2.5 \text{ m})} = 0.58 \frac{\text{m}}{\text{s}^2}$$

$$\text{e) } a = \sqrt{(a_{\text{tan}}^2 + a_R^2)} = \sqrt{\left(0.15 \frac{\text{m}}{\text{s}^2}\right)^2 + \left(0.58 \frac{\text{m}}{\text{s}^2}\right)^2} = 0.60 \text{ m/s}^2$$

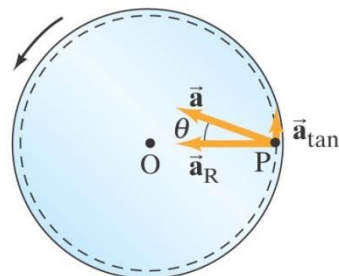
$$\theta = \tan^{-1}\left(\frac{a_{\text{tan}}}{a_R}\right) = \left(0.15 \frac{\text{m}}{\text{s}^2}\right) / \left(0.58 \frac{\text{m}}{\text{s}^2}\right) = 0.25 \text{ rad} \approx 15^\circ$$



(a)

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Giancoli Fig 10.8: Carousel



(b)

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Giancoli Fig 10.8:
Resultant Acceleration

Non-Uniform vs Uniform Circular Motion

We will use the ceiling fan as an example.

When the ceiling fan is stationary, angular velocity $\omega_i = 0$. When the fan is turned on, (assuming that angular acceleration α is constant), the angular velocity of the fan increases to a steady final angular velocity ω_f in time t_1 . Thus, during this period,

$$\omega_f = \omega_i + \alpha t_1.$$

We now stick an object at the tip of one of the blades to focus our attention. The linear (tangential) acceleration a_{tan} of the object (at radius r from the center) is given by $a_{tan} = r\alpha$ and its instantaneous linear (tangential) velocity $v = r\omega$ where ω is the angular velocity at that instant of time t .

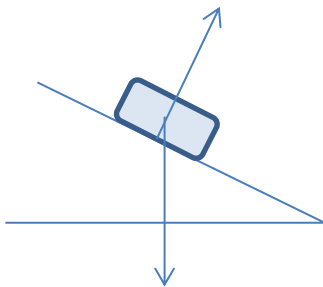
Similarly, the centripetal acceleration at any time is given by $a_r = \frac{v^2}{r} = r\omega^2$.

When the speed of the fan become constant, ω does not change with time and $\alpha = 0$, hence $a_{tan} = 0$. However, $v \neq 0$ and constant in magnitude, hence $a_r = \frac{v^2}{r} = r\omega^2 \neq 0$ and also constant in magnitude. However, vectorially, v and a_r are always changing though the magnitude is constant.

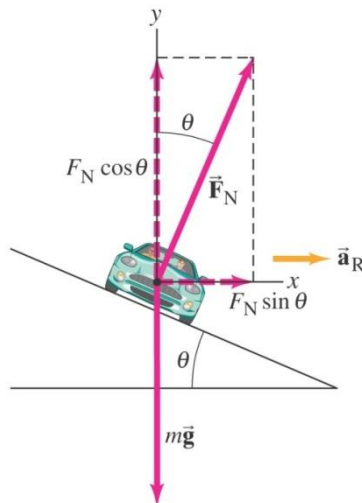


Comparing

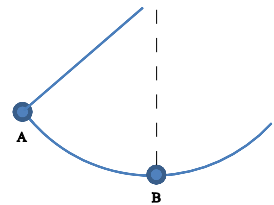
Sliding down slope



Banking round
In horizontal circle



Pendulum swing



"I've been noticing gravity since I was very young."

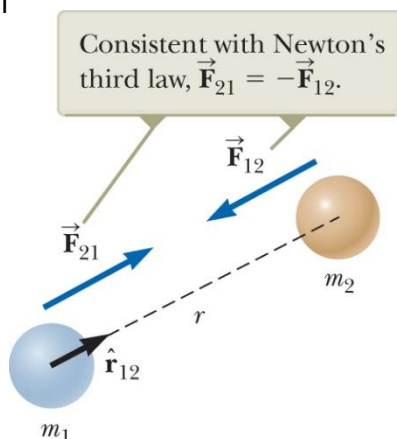
- Cameron Diaz

"One story Einstein liked to tell about his childhood was of a wonder he saw when he was four or five years old: a magnetic compass. The needle's invariable northward swing, guided by an invisible force, profoundly impressed the child. The compass convinced him that there had to be "something behind things, something deeply hidden."

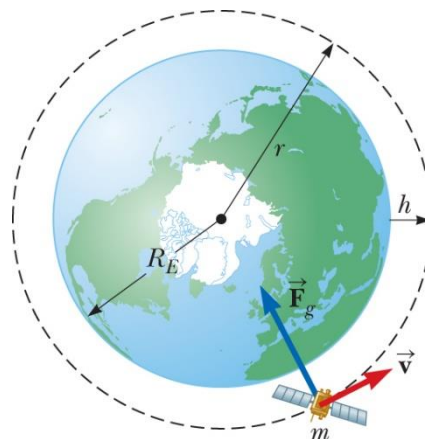
About Albert Einstein

Self-check – after completing the lessons (lecture and tutorial) for this topic, are you able to

1. state Newton's law of gravitation and apply it solve problems in Physics
2. analyze objects in circular orbit under the influence of gravitational force
3. calculate gravitational forces and fields in problems involving one or more massive objects
4. demonstrate understanding of the concept of gravitational force, field and flux through basic calculations
5. use the concept of work done to derive the formula for gravitational potential energy where gravitational field is not constant and solve problems involving motion through large vertical distances where g is not constant
6. state and explain the relationship between gravitational force, field, flux, potential energy and potential



Serway Fig 13.2: Newton's Law of Gravitation

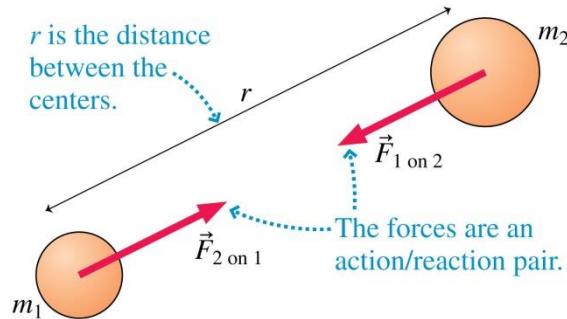


Serway Fig 13.9: Satellite

Gravity and Newton's Law of Gravitation

Newton's Law of gravitation:

Every particle in the universe attracts every other particle with a force that is proportional to the product of their masses and inversely proportional to the square of the distance between them. This force acts along the line joining the two particles.



Knight Fig 6.21: Gravitational force of attraction between two masses

The magnitude of the gravitational force can be written as

$$F_G = \frac{Gm_1m_2}{r^2}$$

where m_1 and m_2 are the masses of the two particles and r is the distance between them, and G is a universal constant which must be measured experimentally and has the same numerical value for all objects. *The force is inversely proportional to the square of the distance between the objects.*

The value of $G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}$. Note here that Big "G" is universal and do not change with location. Small g changes with location and is 9.8 m/s^2 near surface of Earth.

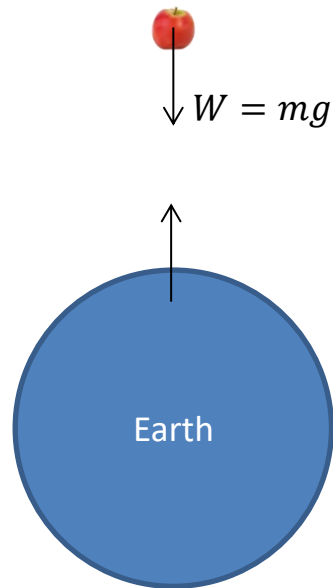
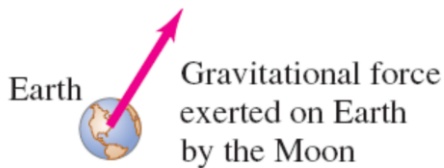
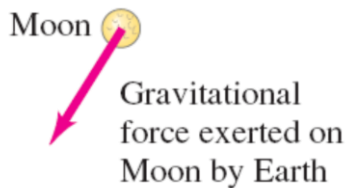
Giancoli pg 141, Example 6-1: Can you attract another person gravitationally?

A 50-kg person and a 70-kg person are sitting on a bench close to each other. Estimate the magnitude of the gravitational force each exerts on the other.

Newton and Apple

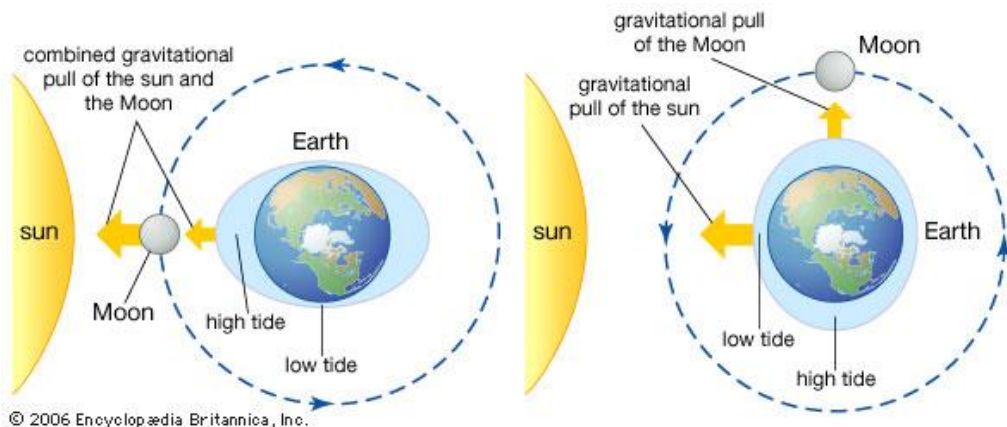


<https://www.flickr.com/photos/hikimgartist/6217869031/>



Giancoli Fig 6.2 The gravitational force one object exerts on a second object is directed toward the first object, and is equal and opposite to the force exerted by the second on the first.

Gravity and Tides



<http://kids.britannica.com/elementary/art-119164/Both-the-sun-and-the-Moon-pull-on-Earths-water>

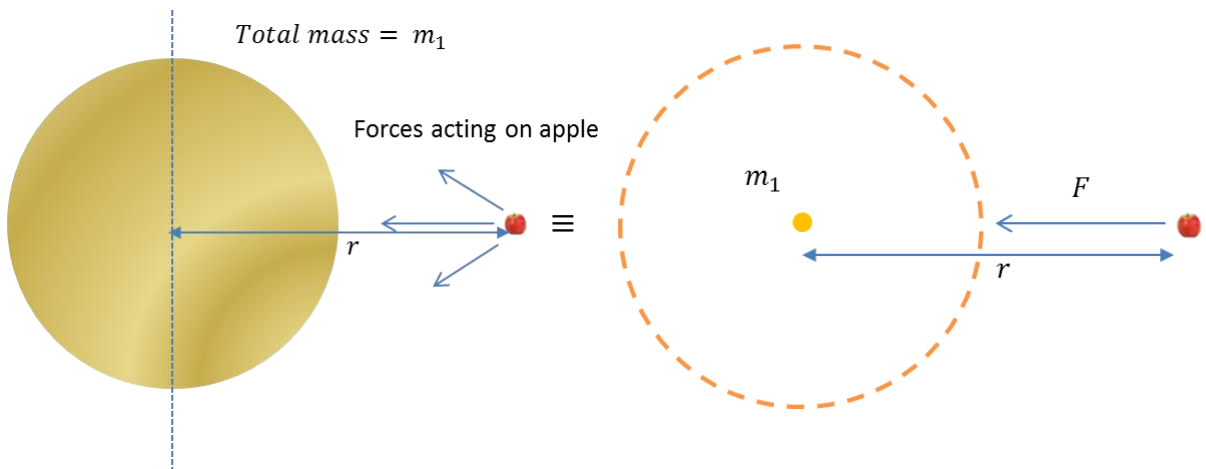
Newton's Law of Gravitation – a little problem



$$F = \frac{Gm_1m_2}{r^2}$$

What is the value of r for this case?

<https://www.flickr.com/photos/hikimgartist/6217869031/>

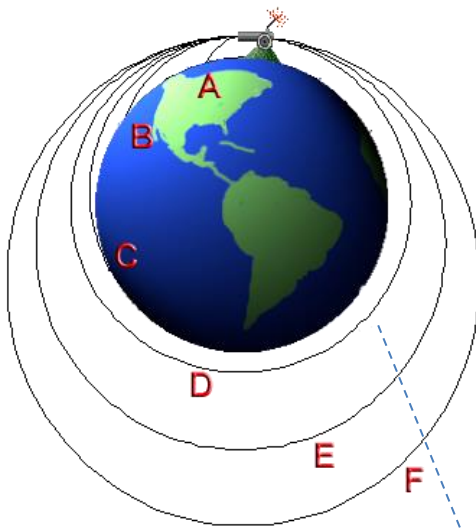
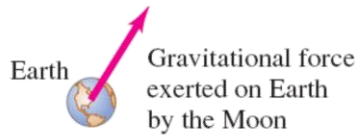
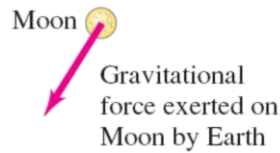


The total force exerted on the apple due to Earth is equivalent to the force due to all mass of Earth concentrated in its own center. (See Appendix D in e-text)

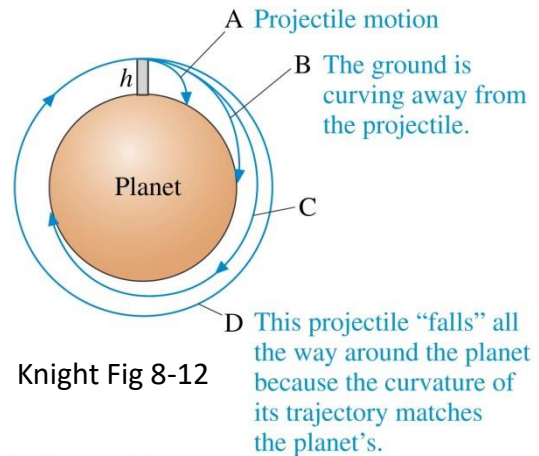
Example [G6.51] Modified

How far above the Earth's surface will the gravitational force on an object be halved as compared to the surface?

Why does the moon not crash into Earth?

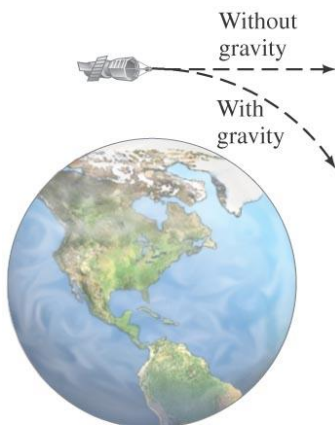


The cannon ball is kept in orbit by its speed—it is continually falling, but the Earth curves from underneath it.



Knight Fig 8-12

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Force along radial direction:

$$F_G = \frac{mv^2}{r}; \quad F_G = mr\omega^2$$

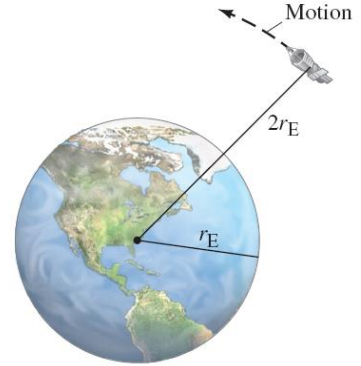
Giancoli pg 146 Fig 6.12:
Gravitational force causes the satellite to deviate from a straight line. At the right distance from Earth's centre and velocity, the satellite moves in a circular path.

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Newton's Law of Gravitation

Giancoli pg 142, An extension of Example 6-2: Spacecraft at $2r_E$.

What is the force of gravity acting on a 2000-kg spacecraft when it orbits two Earth radii from the Earth's center (that is, a distance $r_E = 6380$ km above the Earth's surface)? The mass of the Earth is $m_E = 5.98 \times 10^{24}$ kg.



If the spacecraft is orbiting in a circle at constant speed, what is the centripetal force it experiences?

To stay in orbit, what speed must the spacecraft be moving? How long does it take to complete one round?

If we wish to have a satellite orbiting with a period of 24 hr, how far from the Earth's center should the satellite be? At what speed should it be moving in the orbit? Is energy needed to keep it in motion?

Satellites in Orbits

Giancoli pg 147, Example 6-6: Geosynchronous satellite.

A geosynchronous satellite is one that stays above the same point on the Earth, which is possible only if it is above a point on the equator. Such satellites are used for TV and radio transmission, for weather forecasting, and as communication relays. Determine (a) the height above the Earth's surface such a satellite must orbit, and (b) such a satellite's speed.

$$\frac{Gm_{sat}m_E}{r^2} = m_{sat}r\omega^2 \text{ or } \frac{m_{sat}v^2}{r}$$

(a) Using $m_{sat}r\omega^2$,

$$\frac{Gm_{sat}m_E}{r^2} = m_{sat}r\left(\frac{2\pi}{T}\right)^2$$

$$Gm_E = r^3\left(\frac{2\pi}{T}\right)^2 \Rightarrow$$

$$r^3 = \frac{Gm_ET^2}{4\pi^2} = \frac{\left(6.67 \times 10^{-11} \frac{Nm^2}{kg^2}\right)(5.98 \times 10^{24} kg)(24 \times 3600s)^2}{4\pi^2}$$

$$r = 4.23 \times 10^7 m \text{ or } 42\,300 \text{ km from Earth's center} \\ \Rightarrow 36,000 \text{ km above Earth's surface.}$$

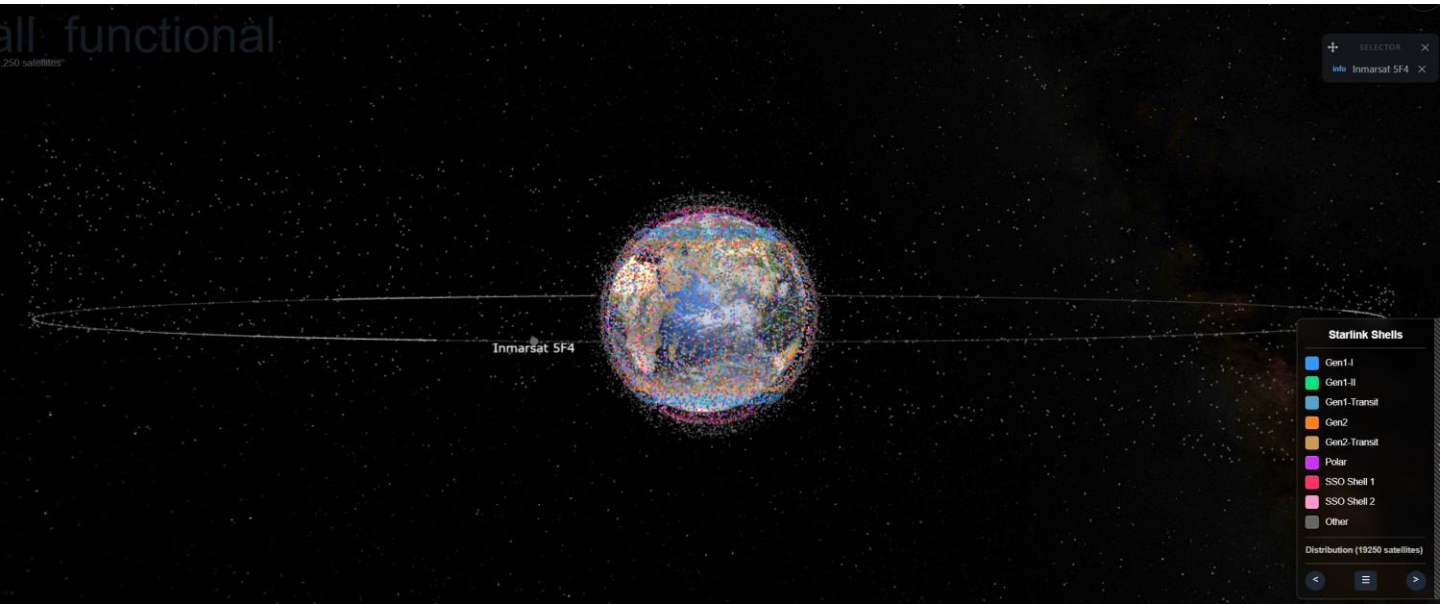
(b) Using $\frac{m_{sat}v^2}{r}$,

$$\frac{Gm_{sat}m_E}{r^2} = \frac{m_{sat}v^2}{r}$$



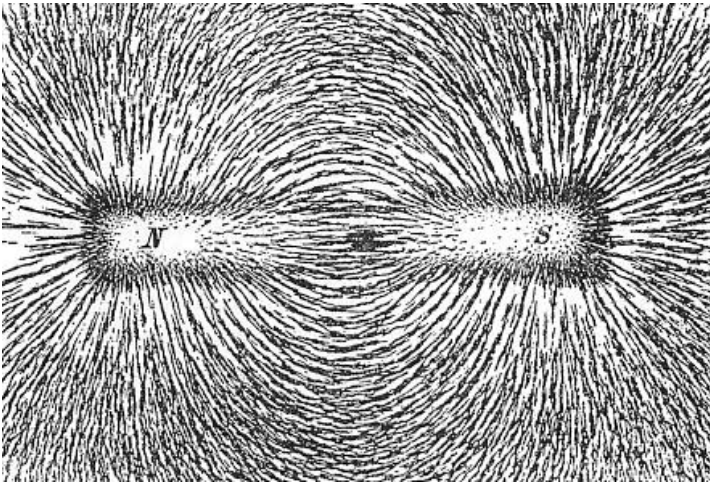
$$v = \sqrt{\frac{Gm_E}{r}} = \sqrt{\frac{\left(6.67 \times 10^{-11} \frac{Nm^2}{kg^2}\right)(5.98 \times 10^{24} kg)}{4.23 \times 10^7 m}} = 3070 \text{ m/s}$$

<https://satellitemap.space/>



Select a satellite and click on the “info”. Using the information on the semi-major axis as “radius”, (check that the orbit looks more circular than elliptical), calculate the period and velocity and tally, check if it agrees with the information provided. Also find out more about what the function of the satellite you selected.

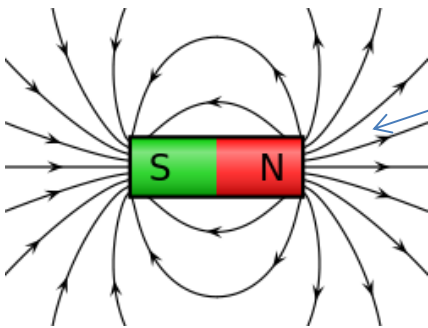
Concept of field



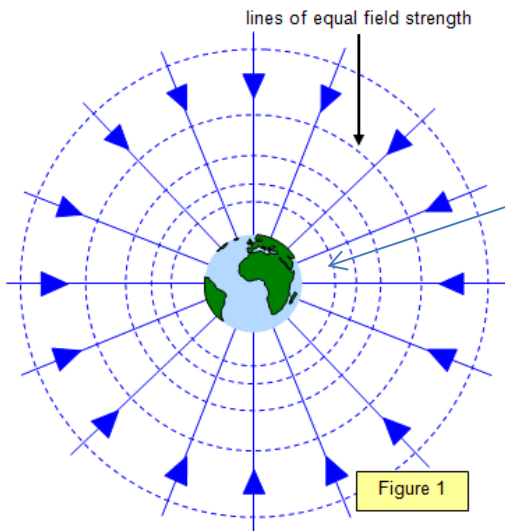
Probably your first encounter... magnetic fields.

http://en.wikipedia.org/wiki/Magnetic_field#/media/File:Magnet0873.png

http://en.wikipedia.org/wiki/Magnetic_field#/media/File:VFPT_cylindrical_magnet_thumb.svg



Closer field lines →
Greater field strength → Greater force



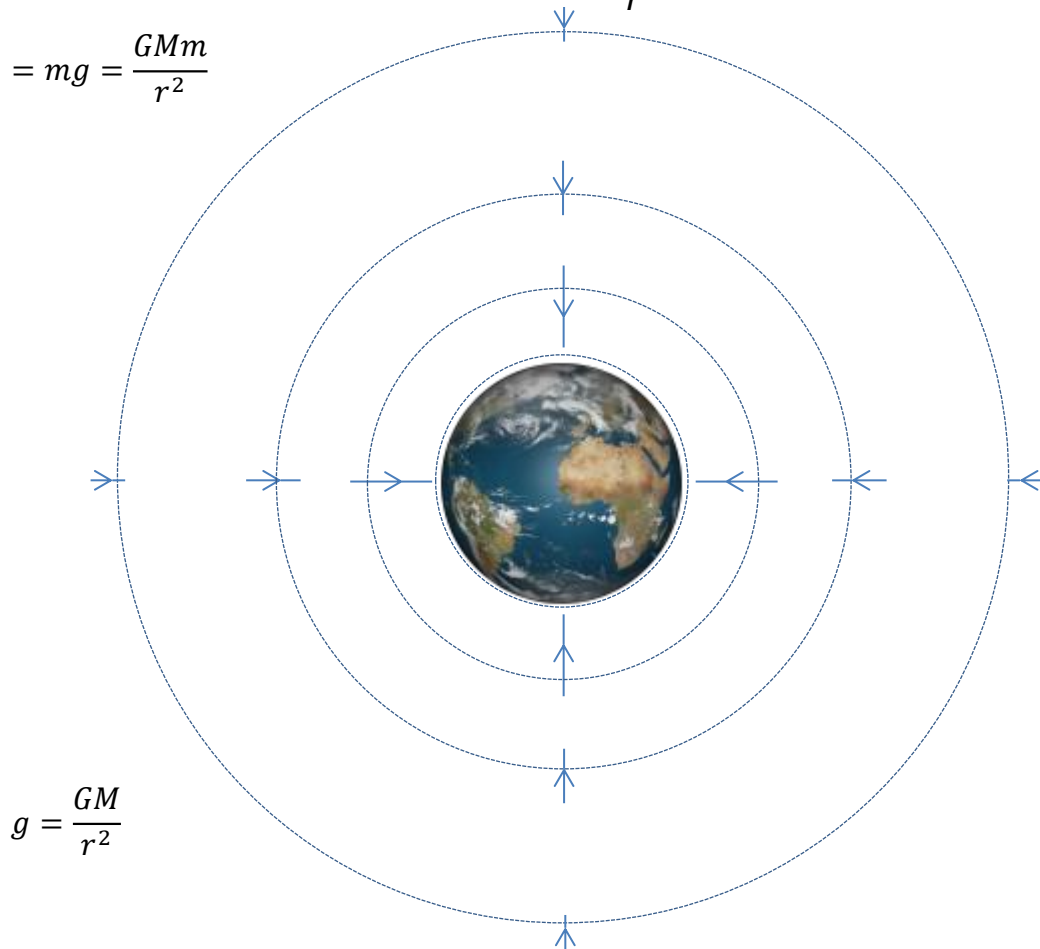
In gravitation,

Larger field lines density →
Greater field strength → Greater force

Gravitational Field Strength

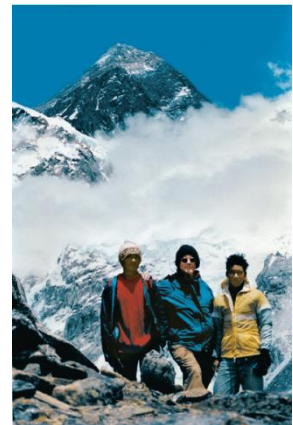
Gravitational force per unit mass. $g = \frac{GM}{r^2}$

$$F_g = mg = \frac{GMm}{r^2}$$



More Examples

Giancoli Example 6-4: Gravity on Everest. Estimate the effective value of g on the top of Mt. Everest, 8850 m above sea level. That is, what is the acceleration due to gravity at this altitude? ($r_E = 6380$ km, $M_E = 5.98 \times 10^{24}$ kg)



Google home: Oil exploration, gravimeter

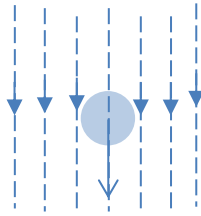
EXERCISE

1. [G6.2] Calculate the acceleration due to gravity on the Moon. The Moon's radius is $1.74 \times 10^6 m$ and its mass is $7.35 \times 10^{22} kg$.
2. The gravitational field at the surface of the Earth of radius r_E is $g = 9.8 \text{ N/kg}$. Determine the gravitational field (or acceleration of free-fall) at twice of Earth's radius, $2R_E$ and sketch a graph of how g varies with distance from the center of the Earth, starting from the surface of Earth.

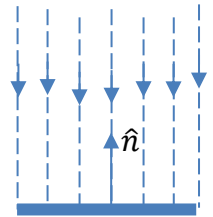
Force, Field and Flux



Force



Field



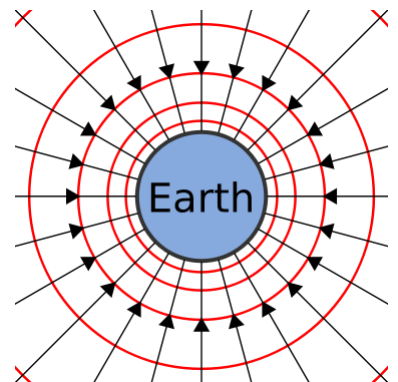
Area A

Flux

HOSY@ntu.edu.sg

Near the surface of Earth, the gravitational field is $g = 9.8 \text{ N/kg}$.

- Determine the gravitational flux through a piece of paper 3 m^2 which is placed parallel to the ground level.
- If the paper is tilted such that it is at 30° to the horizontal, determine the flux through the 3 m^2 paper.
- If the radius of Earth is 6400 km , determine the total gravitational flux of Earth.
- How will the total gravitational flux of Earth change if now a measurement is done with an imaginary sphere with radius $2r_E$.



More Generalized Concept of Flux



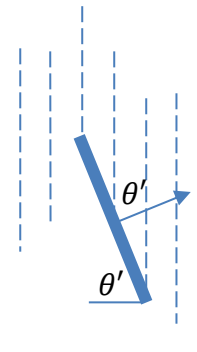
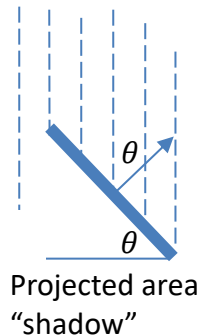
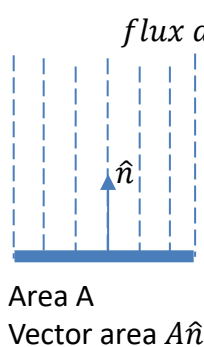
Heavy Rain – “high rain flux per unit area”
(high flux density)



Light Rain – “low rain flux per unit area”
(low flux density)

<http://99fm.com.na/unseasonably-heavy-rain-expected/>

<http://www.chachacharming.com/music/i-like-london-in-the-rain-mix/>



HOSY@ntu.edu.sg

$$\begin{aligned}\text{Total flux “captured” by area } \Phi &= \text{“flux density”} \times \text{Projected Area} \\ &= \text{“flux density”} \times \text{Area} \cos \theta \\ &= A |\vec{f}| |\hat{n}| \cos \theta = \vec{f} \cdot (A\hat{n})\end{aligned}$$

For curved surface, $d\vec{A} = Ad\vec{n}$

Can also think of solar panels.

$$\text{Total flux, } \Phi = \int \vec{f} \cdot d\vec{A}$$

For closed curved surface, (convention, outwards +ve)

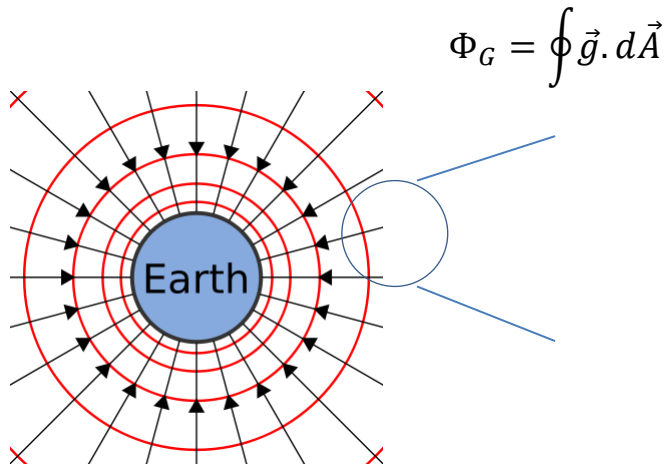
$$\text{Total flux, } \Phi = \oint \vec{f} \cdot d\vec{A}$$

Thus, for gravitational fields, the total gravitational flux Φ_G enclosed by a shell centred with Earth's center and encloses Earth,

$$\Phi_G = \oint \vec{g} \cdot d\vec{A}$$

Gravitational Flux

Thus, for gravitational fields, the total gravitational flux Φ_G enclosed by a shell centred with Earth's center and encloses Earth,



Fortunately for us,

1. $\vec{g} \perp d\vec{A}$ for the whole sphere (for any of the red spheres).
2. For any selected sphere, the magnitude of g is constant on the surface of the sphere.

Thus, the value of the gravitational flux Φ_G can be evaluated easily,

$$\Phi_G =$$

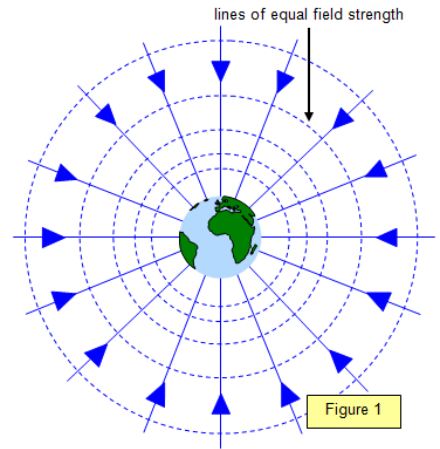
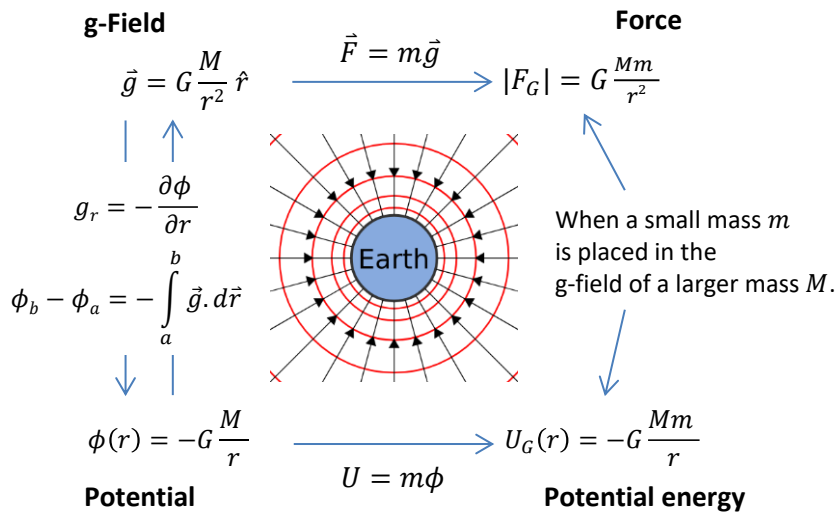
Gauss made an observation, select any “imaginary” surface (known as Gaussian surface) to enclose the whole Earth,

$$\begin{aligned} \text{Total flux } \Phi_G \text{ through area of surface} \\ = \text{constant} \times \text{mass enclosed by surface} \end{aligned}$$

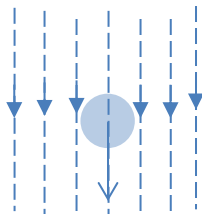
Example: The mass of Mars is $6.39 \times 10^{23} \text{ kg}$ and radius is 3390 km. What is the gravitational flux of Mars $\Phi_{G, \text{Mars}}$?

Summary of Key Ideas (To revisit later)

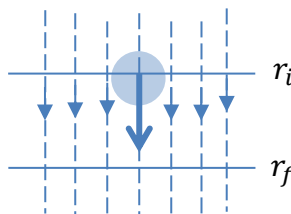
Mass (Radial Fields)



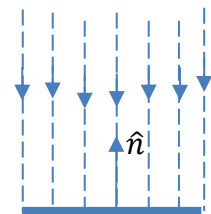
$$\Phi_G = \oint \vec{g} \cdot d\vec{A}$$



Force
& Field



Potential
& Energy



Area A
Total Flux &
Mass / Charge
enclosed

Coming next...

Point Charges (Radial Fields)

